235129: quartz metadiorite, Duketon

(Yilgarn Craton granites, Burtville Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Duketon (SG 51-14), Duketon (3342)
MGA Zone 51, 428613E 6943353N

Sampled on 27 October 2020

This sample was collected from the 50.50 – 51.88 m depth interval of drillcore LSHD7, drilled in 1982 by Western Mining Corporation Limited at their Swanson Hill prospect (Brooke, 1983). The drillhole is located north-northwest of the historic Duketon battery site on Banjawarn Station, about 6.6 km east-southeast of Mason Hill, 5.3 km north of Pinje-Eda Hill, and 0.8 km southeast of the Lake Wells Road.

Geological context

The unit sampled is a quartz metadiorite of the northern Duketon greenstone belt, near the western margin of the Burtville Terrane in the Eastern Goldfields Superterrane (Pawley et al., 2012). Samples of metadacite, collected from drillcore LSHD6, about 4.4 km to the east-southeast, yielded igneous crystallization ages of 2714 ± 5 Ma (GSWA 153317, Nelson, 2004a) and 2719 ± 5 Ma (GSWA 153318, Nelson, 2004b). Metadacite in drillcore LSHD10, about 11.3 km to the north, yielded an igneous crystallization age of 2808 ± 3 Ma (GSWA 235199, Wingate et al., 2021). Near the western margin of the Duketon belt, about 32 km to the south, a felsic volcaniclastic rock provided an igneous crystallization age of 2714 ± 4 Ma, and a porphyritic intrusion within that succession yielded an igneous crystallization age of 2704 ± 6 Ma (samples E380, E382, Kositcin et al., 2008).

Petrographic description

The sample is an equigranular, medium-grained, hornblende–biotite quartz metadiorite (Fig. 1), consisting of about 60% plagioclase, 15–20% hornblende–actinolite, 10–15% quartz, 5–8% biotite, 5% chlorite, and accessory magnetite–titanite, apatite and zircon. Plagioclase (andesine) is subhedral, tabular, randomly oriented, up to 2.5 mm long, and partially replaced by sericite and fine epidote. Green hornblende is subhedral, tabular, up to 2 mm in size, partially replaced by actinolite and chlorite, and occurs in aggregates with biotite and magnetite. Quartz is anhedral, irregular, up to 1.5 mm in size, and interstitial to plagioclase. Quartz is moderately strained, exhibits incipient subgraining and recrystallization around grain margins, and is completely recrystallized in narrow zones, or along fractures. Biotite is forms subhedral, equant laths up to 1 mm long. Pale green chlorite has replaced biotite and to a lesser extent, hornblende, and also occurs in decussate aggregates in fractures, together with recrystallized quartz. Magnetite is anhedral, equant, up to 0.2 mm across, and partially overgrown or replaced by titanite. Apatite is euhedral, prismatic, and up to 0.15 mm long. The narrow zones of recrystallized quartz and chlorite and the presence of strained quartz are interpreted as the effects of deformation at low (greenschist facies) metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless and mainly euhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 25–26 June 2021, using SHRIMP-B. Ten analyses of the M257 standard were obtained during the session, of which seven analyses indicated an external spot-to spot (reproducibility) uncertainty of 1.29% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.53% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Sixteen analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 16 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2716 ± 5 Ma (MSWD = 1.0).

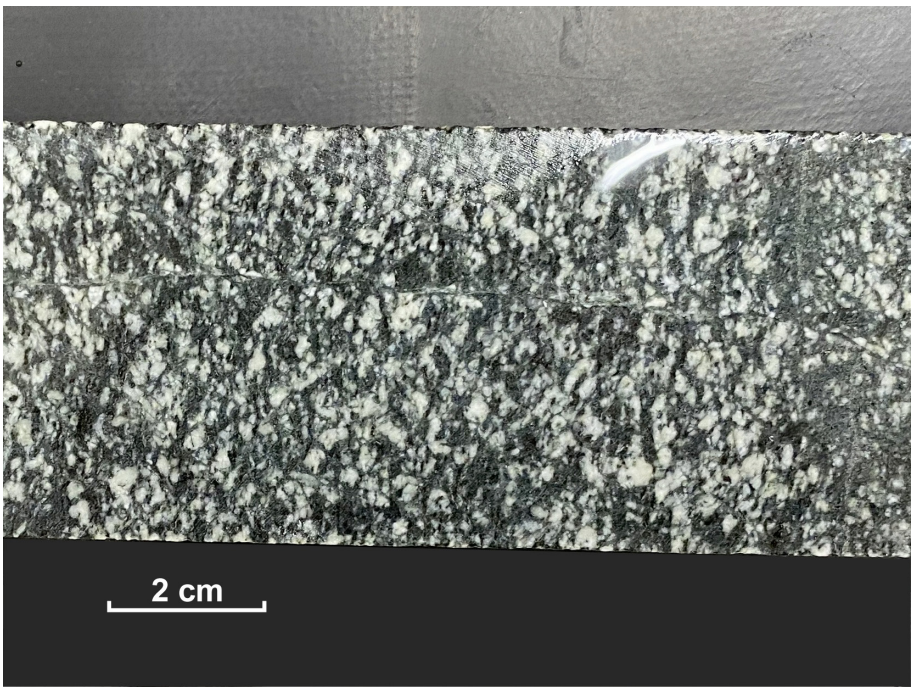

Figure 1. Drillcore image (wet surface) of sample 235129: quartz metadiorite, Duketon

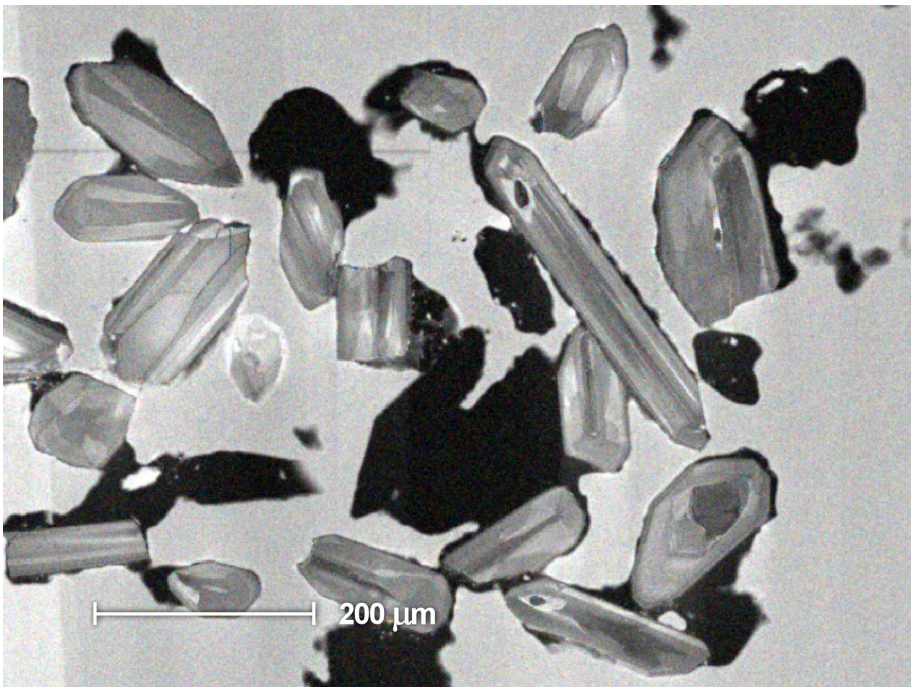

Figure 2. Cathodoluminescence image of representative zircons from sample 235129: quartz metadiorite, Duketon

Table 1. Ion microprobe analytical results for zircons from sample 235129: quartz metadiorite, Duketon

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	9	9.1	70	29	0.42	0.000	1.848	0.035	0.18476	0.00147	1.848	0.035	0.18476	0.00147	2788	44	2696	13	-3.4
I	4	4.1	93	51	0.55	0.085	1.933	0.034	0.18627	0.00125	1.935	0.034	0.18551	0.00131	2686	40	2703	12	0.6
I	8	8.1	71	44	0.62	0.173	1.968	0.037	0.18730	0.00146	1.972	0.037	0.18576	0.00159	2645	41	2705	14	2.2
I	7	7.1	117	85	0.72	0.000	1.897	0.032	0.18592	0.00112	1.897	0.032	0.18592	0.00112	2729	39	2706	10	-0.8
I	11	11.1	170	86	0.51	0.000	1.912	0.031	0.18610	0.00091	1.912	0.031	0.18610	0.00091	2712	37	2708	8	-0.1
I	15	15.1	186	112	0.60	0.022	1.886	0.030	0.18657	0.00089	1.887	0.030	0.18637	0.00090	2741	37	2710	8	-1.1
I	3	3.1	121	91	0.75	0.016	1.906	0.032	0.18689	0.00110	1.907	0.032	0.18674	0.00110	2718	38	2714	10	-0.2
I	13	13.1	285	245	0.86	0.007	1.927	0.030	0.18693	0.00073	1.927	0.030	0.18686	0.00073	2695	35	2715	6	0.7
I	2	2.1	129	93	0.72	0.047	1.982	0.033	0.18735	0.00110	1.983	0.033	0.18693	0.00113	2632	37	2715	10	3.1
I	12	12.1	117	48	0.41	-0.035	2.002	0.034	0.18675	0.00115	2.001	0.034	0.18706	0.00117	2613	37	2717	10	3.8
I	16	16.1	103	78	0.76	-0.112	1.875	0.033	0.18615	0.00118	1.873	0.033	0.18715	0.00125	2758	40	2717	11	-1.5
I	14	14.1	148	80	0.55	-0.073	1.921	0.032	0.18652	0.00104	1.919	0.032	0.18716	0.00107	2704	38	2717	9	0.5
I	10	10.1	147	99	0.67	0.040	1.931	0.032	0.18762	0.00100	1.932	0.032	0.18726	0.00102	2689	37	2718	9	1.1
I	1	1.1	162	85	0.53	-0.023	1.931	0.032	0.18751	0.00092	1.930	0.032	0.18771	0.00093	2691	38	2722	8	1.1
I	5	5.1	108	83	0.77	0.054	1.974	0.034	0.18946	0.00117	1.975	0.034	0.18898	0.00120	2641	38	2733	10	3.4
I	6	6.1	118	90	0.76	-0.063	1.933	0.033	0.18900	0.00109	1.932	0.033	0.18956	0.00113	2689	38	2738	10	1.8

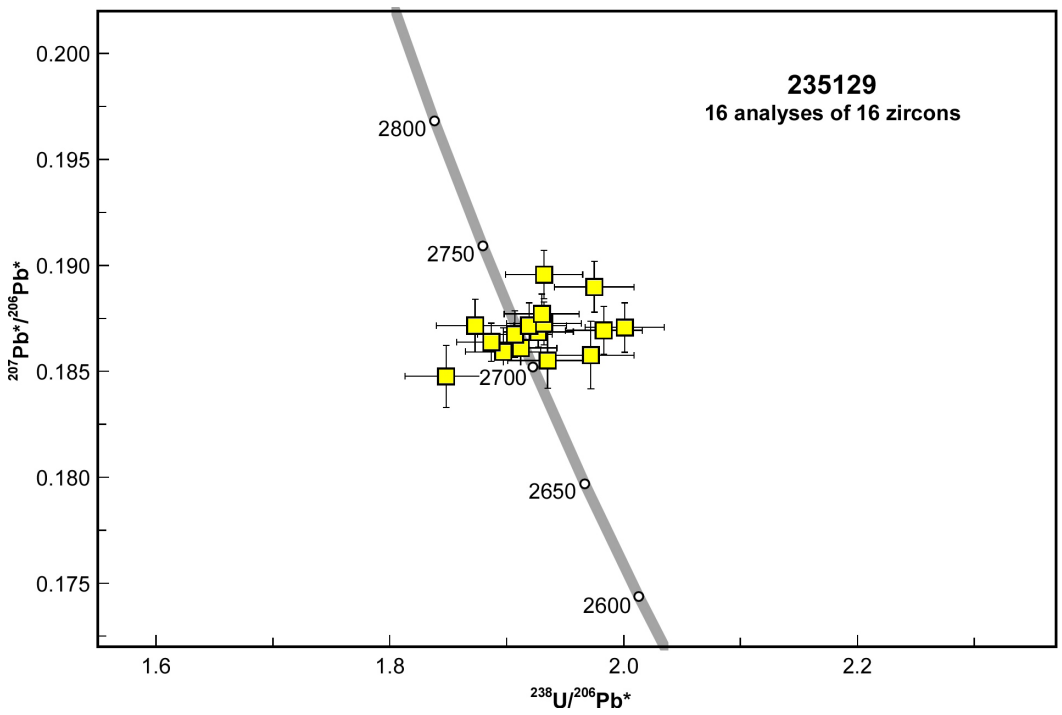

Figure 3. U–Pb analytical data for sample 235129: quartz metadiorite, Duketon. Yellow squares indicate Group I (magmatic zircons)

The date of 2716 ± 5 Ma for the 16 analyses in Group I is interpreted as the igneous crystallization age of the quartz diorite.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Grech, LL 2021, 235129: quartz metadiorite, Duketon; Geochronology Record 1833: Geological Survey of Western Australia, 4p.

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